

6.1

kako se g spreminja z višino

$$F_g = \frac{GmM}{r^2} = mg(r)$$

$$g_0 = \frac{MG}{R_z^2} \rightarrow MG = g_0 R_z^2$$

$$g(r_z + h) = \frac{MG}{(r_z + h)^2}$$

$$g(h) = g_0 \left(\frac{R_z}{R_z + h} \right)^2 = g_0 \left(\frac{1}{1 + \frac{h}{R_z}} \right)^2 = g_0 \frac{1}{1 + \frac{2h}{R_z} + \left(\frac{h}{R_z}\right)^2} = g_0 \frac{1}{1 + \frac{2h}{R_z}} =$$

$$= g_0 \left(1 - \frac{2h}{R_z} \right), \quad h \text{ majhen}$$

6.2

Kako pridemo iz $W_{gr,p} = -\frac{GM_m}{r}$ v $W_p = mgh$

$$W_{gr} = -\frac{GM_m}{r} = -\frac{GM_m \cdot R_z}{(R_z + h) \cdot R_z} = -g_0 \cdot m \frac{R_z}{1 + \frac{h}{R_z}} = \underbrace{-g_0 R_z m}_{\text{konst.}} + g_0 h m$$

konst., če gledamo ΔW_p , je ok.

Taylorjev razvoj:

$$\text{-v okolici } x_0: f(x) = f(x_0) + \frac{f'(x_0)(x-x_0)^1}{1!} + \dots + \frac{f^{(n)}(x_0)(x-x_0)^n}{n!}$$

$$f(x) = \frac{1}{1+x} \doteq 1-x$$

$$\frac{h}{R_z} = x \doteq 0$$

29/28

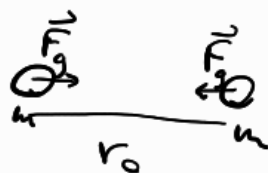
$$m = 10g$$

$$r_0 = 10cm$$

$$t = 2min$$

$$v(t) = ?$$

$$or = ?$$



$$\int dt$$


$$F_g = G \frac{m_1 m_2}{r^2} = m a$$

$$a = \frac{Gm}{r^2} \rightarrow a_0 = \frac{6,67 \cdot 10^{-11} \cdot 10^{-2}}{10^{-2}} =$$

$$= 6,67 \cdot 10^{-11} \frac{m}{s^2}$$

$$\Downarrow$$

$$a \approx konst.$$

napaka:

$$= \frac{F_0 - F}{F_0} = 1 - \frac{m a_0 G m r_0^2}{m a_0 r^2 G m} = 1 - \left(\frac{r_0}{r_0 - \Delta r} \right)^2$$

$$\frac{\Delta r}{2} = \frac{a t^2}{2}$$

$$\Delta r = 9,6 \cdot 10^{-7} m$$

$$= 1 - \left(\frac{1}{1 - \frac{\Delta r}{r_0}} \right)^2 \approx 1 - \frac{1}{1 - \frac{2 \Delta r}{r_0}} = 1 - \left(1 + \frac{2 \Delta r}{r_0} \right) = - \frac{2 \Delta r}{r_0} \approx 2 \cdot 10^{-5}$$

rel. nap.

29/39

$$M = 10t$$

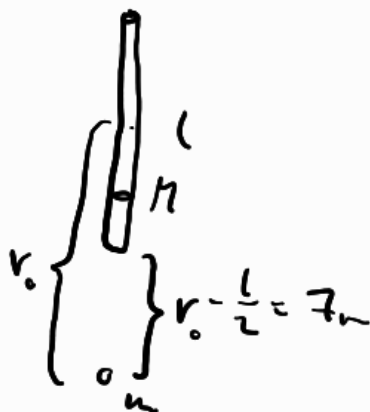
$$l = 10m$$

$$m = 1t$$

$$r = 12m$$

$$a_m = ?$$

$$a_M = ?$$



$$dF = G m \frac{dM}{r^2} \quad dM = M \frac{dr}{l}$$

$$F = \int_{r_0 - \frac{l}{2}}^{r_0 + \frac{l}{2}} G m M \frac{dr}{r^2} = -G m M \frac{1}{r} \left(\frac{1}{r} \right) \Big|_{r_0 - \frac{l}{2}}^{r_0 + \frac{l}{2}}$$

$$F = \frac{G m M}{l} \left(\frac{1}{r_0 - \frac{l}{2}} - \frac{1}{r_0 + \frac{l}{2}} \right) =$$

$$= \frac{G m M}{l} \left(\frac{4 \cdot 1}{r_0^2 - \frac{l^2}{4}} \right) = \frac{G m M}{r_0^2 - \frac{l^2}{4}}$$

$$a_M = \frac{F}{M} = \frac{G m}{r_0^2 - \frac{l^2}{4}} = \frac{6,67 \cdot 10^{-11} N \cdot 10^3 kg}{kg^2 (119 m^2)} = 5,6 \cdot 10^{-10} \frac{m}{s^2}$$

$$a_m = \frac{F}{m} = \frac{G M}{r_0^2 - \frac{l^2}{4}} = \frac{6,67 \cdot 10^{-11} N \cdot 10^4 kg}{119 m^2 kg} = 5,6 \cdot 10^{-9} \frac{m}{s^2}$$

29/30

$$F_x = 0$$

$$M = 200t$$

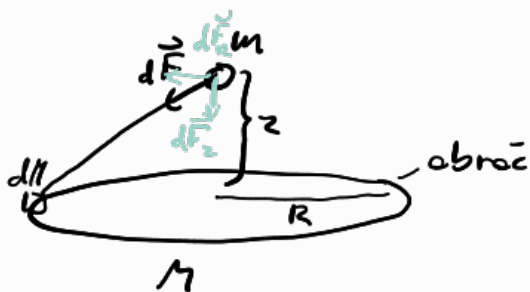
$$R = 12m$$

$$z = 10m$$

$$m = 20t$$

$$a_m = ?$$

$$a_M = ?$$



$$dF = G_m \frac{dM}{R^2 + z^2}$$

$$\frac{dF_z}{dF} = \frac{z}{\sqrt{R^2 + z^2}}$$

$$dF_z = \frac{G_m z dM}{(R^2 + z^2)^{3/2}}$$

$$a_m = 3,5 \cdot 10^{-8} \frac{m}{s^2}$$

$$a_M = 3,5 \cdot 10^{-9} \frac{m}{s^2}$$

$$F_z = \int_0^{\pi} \frac{G_m z dM}{(R^2 + z^2)^{3/2}} = \frac{G_m M z}{(R^2 + z^2)^{3/2}} =$$
$$= \frac{6,67 \cdot 10^{-11} \cdot 20 \cdot 10^3 \cdot 200 \cdot 10^3 \cdot 10}{(244)^{3/2}}$$

Lažje:

$$W_p = - \frac{G_m M}{\sqrt{R^2 + z^2}}$$

$$\vec{F} = - \vec{\nabla} W_p = - \left(\frac{\partial W_p}{\partial x}, \frac{\partial W_p}{\partial y}, \frac{\partial W_p}{\partial z} \right) = - \left(0, 0, - \frac{G_m M \cdot z}{(R^2 + z^2)^{3/2}} \cdot \frac{z}{z} \right) =$$
$$= \frac{G_m M z}{(R^2 + z^2)^{3/2}}$$

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